RACE
Regular Analysis Through Continuous Exercise

Ionic Equilibrium : RACE-01

Ionic Equilibrium in Solutions; Arrhenius Concept of Acids and Bases, The Bronsted-Lowry Acids and Bases, Lewis Acids and Bases

1. In the following reaction

$$\underset{A}{[Al(H_2O)_6]^{3+}} + \underset{B}{H_2O} \rightleftharpoons \underset{C}{H_3O^+} + \underset{D}{[Al(H_2O)_5OH]^{+2}}$$

(A) A is an acid, B is a base
(B) A is a base, B is and acid
(C) C is conjugate acid of B and D is conjugate base of A
(D) C is conjugate base of B and D is conjugate acid of A

CIE288

2. Find the number of aqueous solution which turn litmus paper blue
(i) NH_4OH (ii) NaOH (iii) NH_4Cl (iv) CH_3COONa
(v) $NaHSO_4$ (vi) HCl

CIE269

Electrolytes; Ionization Constant of Water; Ionic Product of Water (Kw), Ionization of Acids and Bases

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3. The dissociation of water at 25°C is 1.9×10^{-7} percent and the density of water is 1.0 g/cm^3. The self-ionization constant of water is :
(A) 1.8×10^{-16} (B) 3.42×10^{-8} (C) 1.00×10^{-14} (D) 2.00×10^{-16}

CIE276

The pH scale; Calculation of pH of solution containing Strong Acid and Strong Base; Dilution case

4. The pH of solution having doubled the alkaline concentration of water at 25°C is :
(A) 14 (B) 0 (C) 8.2 (D) 7.3

CIE272

5. For pure water
(A) pH increases with increase in temperature
(B) pH decreases with increase in temperature
(C) pH = 7 at temperature of 25°C
(D) pH increases at low temperature but decreases at high temperature

CIE280

6. 298 K pure T_2O has pT (like pH) is 7.60. Find out the pT of a solution prepared by adding 10 ml of 0.2 M TCl to 15 ml of 0.20 M NaOT. (Given log 2 = 0.30)

First multiply your answer with 100. Then fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

CIE290

7. For pure water on increasing temperature
(A) pH decreases (B) pOH decreases
(C) pH increases (D) pOH increases

CIE287

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Calculation of pH of solution of acid-base mixtures containing weak acids and weak bases

8. What would be the pH of an ammonia solution if that of an acetic acid solution of equal strength is 3.2 ? Assume dissociation constant for NH_3 & acetic acid are equal.

(A) 3.2 (B) 11.2 (C) 9.6 (D) 10.8

CIE291

9. The pK_a of acetyl salicylic acid (aspirin) is 3.5. The pH of gastric juice in human stomach is about 2–3 and pH in the small intestine is about 8. Aspirin will be :

(A) Unionised in the small intestine and in the stomach

(B) Completely ionised in the small intestine and in the stomach

(C) Ionised in the stomach and almost unionised in the small intestine.

(D) Ionised in the small intestine and almost unionised in the stomach.

CIE261

10. The position of equilibrium lies to right in each of following cases:

$AH^+ + B \rightleftharpoons A + BH^+$ ($K_{eq} = 10^4$)

$B + C - H \rightleftharpoons C^- + BH^+$ ($K_{eq} = 10^6$)

$A + C - H \rightleftharpoons C^- + AH^+$ ($K_{eq} = 10^3$)

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Based on this information, what is the order of acid strength.

(A) C–H > BH^+ > AH^+ > B (B) C–H > BH^+ > AH^+ > B

(C) C–H > AH^+ > BH^+ > B (D) AH^+ > C–H > B > BH^+

CIE264

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	AC	3	C	D	BCD	13.80	AB	D	D	C

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Ionic Equilibrium : RACE-02

Calculation of pH of solution of acid-base mixtures containing weak acids and weak bases

1. What is the $[HCOO^-]$ in the solution that contains 0.015 M HCOOH and 0.02 M HCl?

Ka HCOOH = 1.8×10^{-4}

(A) 1.8×10^{-4} (B) 1.35×10^{-4} (C) 1.8×10^{-2} (D) 8×10^{-3}

CIE274

2. Questions given below consist of two statements each printed as Assertion (A) and Reason (R); while answering these questions you are required of choose any one of the following four responses :

Assertion : 10^{-4} M HCl solution is more acidic compared to 1M HCOOH solution. ($K_a = 10^{-3}$ M)

Reason : HCl is strong acid compared to HCOOH.

(A) If both (A) and (R) are true and (R) is the correct explanation of (A)

(B) If both (A) and (R) are true but (R) is not correct explanation of (A)

(C) If (A) is true but (R) is false

(D) If (A) is false and (R) is true

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CIE281

3. What is the degree of dissociation of weak acid HA (C = 0.1 M) in presence of strong acid HB (C = 0.1 M). Given : K_a(weak acid) = 10^{-6}

(A) 10^{-5} (B) 10^{-6} (C) 10^{-4} (D) 10^{-3}

CIE284

Calculation of pH of solution containing Di- and polybasic acids and Di-and polyacidic bases

4. Calculate the pH of 6.66×10^{-3} M solution of $Al(OH)_3$, it's first dissociation is 100% where as second dissociation is 50% and third dissociation is negligible at 25°C.

(A) 2 (B) 12 (C) 11 (D) 3

CIE262

5. Three solution of strong electrolytes 25 ml of 0.1 M HX, 25, ml of 0.1 M H_2Y and 50 ml of 0.1N $Z(OH)_2$ are mixed. What is the pOH of the solution? [Given log 5 = 0.7]

(A) 1.6 (B) 7 (C) 11.6 (D) 12.4

CIE297

6. Approximate pH of 0.10 M aqueous H_2S solution having K_1 and K_2 for H_2S at 25°C 10^{-7} and 10^{-13} respectively is :

(A) 4 (B) 5 (C) 9 (D) 8

CIE277

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7. 100 mL of 0.5 M hydrazoic acid (N_3H Ka = 3.6×10^{-4} and 400 mL of 0.1 M cyanic acid (HOCN, Ka = 8×10^{-4}) are mixed. Which of the following is true for final solution?

(A) $[H^+] = 2 \times 10^{-2}$ M (B) $[N^{3-}] = 3.6 \times 10^{-2}$ M

(C) $[OCN^-] = 4.571 \times 10^{-3}$ M (D) $[OCN^-] = 6.4 \times 10^{-3}$ M

CIE278

8. Calculate the pH at which the following reaction will be at equilibrium in basic medium :

$A_2(s) \rightleftharpoons A^-(aq.) + AO_3^-(aq.)$

when the equilibrium concentration at 300K are -

$[A^-] = 0.1M$, $[AO_3^-] = 0.1$ M

Given $\Delta G°_f$ of $OH^-_{(aq)} = -150$ kJ/mole,

$\Delta G°_f$ of $H_2O_{(l)} = -233$ kJ/mole,

$\Delta G°_f$ of $A^-_{(aq)} = -50$ kJ/mole,

$\Delta G°_f$ of $AO_3^-{}_{(aq)} = -123.5$ kJ/mole,

$R = \frac{25}{3} J mole^{-1}, K^{-1}$, $\log_e 10 = 2.3$

CIE300

Salt and types of salts; Hydrolysis of Salts and the pH of their Solutions; Monovalent Salts

9. Which of the following has maximum pH

(A) H_2O at 0°C ($K_w = 10^{-16} M^2$)

(B) 0.1M NH_4Cl (at 25°C)

(C) 0.1M CH_3COONH_4 (at 25°C) [Given : $(K_a)_{CH_3COOH} = (K_b)_{NH_4OH}$]

(D) 0.1 M CH_3COOH (at 25°C) [$(K_b)_{CH_3COO^-} = 10^{-9}$M]

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CIE293

10. Ka for CH_3COOH is 1.8×10^{-5} and K_b for NH_4OH is 1.8×10^{-5}. The pH of ammonium acetate will be

(A) 7.005 (B) 4.75

(C) 7.0 (D) Between 6 and 7

CIE296

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	D	A	B	D	A	D	8	A	C

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Ionic Equilibrium : RACE-03

Salt and types of salts; Hydrolysis of Salts and the pH of their Solutions; Monovalent Salts

1. **Assertion (A) :** Aqueous solution of ammonium carbonate is basic.
Reason (R) : Acidic/basic nature of a salt solution of a salt of weak acid and weak base depends on K_a and K_b value of the acid and the base forming it.
(A) Both A and R are true and R is correct explanation of A.
(B) Both A and R are true but R is not correct explanation of A.
(C) A is true but R is false.
(D) Both A and R are false.

CIE263

2. The correct order of increasing $[H_3O^+]$ in the following aqueous solution is
(A) 0.01 M H_2S < 0.01 M H_2SO_4 < 0.01 M NaCl < 0.01 M $NaNO_2$
(B) 0.01 M NaCl < 0.01 M $NaNO_2$ < 0.01 M H_2S < 0.01 M H_2SO_4
(C) 0.01 M $NaNO_2$ < 0.01 M NaCl < 0.01 M H_2S < 0.01 M H_2SO_4
(D) 0.01 M H_2S < 0.01 M $NaNO_2$ < 0.01 M NaCl < 0.01 M H_2SO_4

CIE273

3. When a salt of weak acid and weak base is dissolved in water, the pH of the resulting solution will be
(A) 7 (B) Depends on the value of K_a and K_b
(C) Less than 7 (D) Greater than 7

CIE279

4. What is the aq. ammonia concentration of a solution prepared by dissolving 0.15 mole of $NH_4^+CH_3COO^-$ in 1 L H_2O. $\left[K_{a(CH_3COOH)} = 1.8\times10^{-5}; K_{b(NH_4OH)} = 1.8\times10^{-5}\right]$
(A) 8.3×10^{-4} (B) 0.15 (C) 6.4×10^{-4} (D) 3.8×10^{-4}

CIE285

Buffer Solution; Designing of Buffer solution; pH of Buffer

5. What volume of 0.2 M RNH_3Cl solution in ml should be added to 100 ml of 0.1 M RNH_2 solution to produce a buffer solution of pH = 8.7 at 25°C ?
Given pK_b of RNH_2 = 5 ; log 2 = 0.3
(A) 50 ml (B) 100 ml (C) 200 ml (D) None of these

CIE271

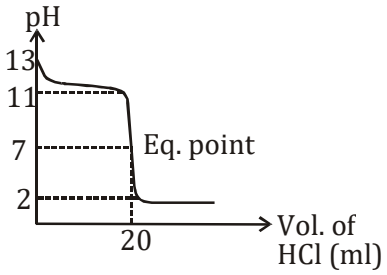
pH Curve

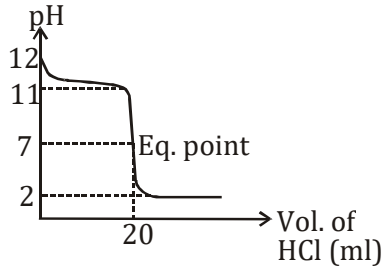
6. 10 ml of 0.1 M tribasic acid H_3A is titrated with 0.1 M NaOH solution. What is the ratio of $\frac{[H_3A]}{[A^{3-}]}$ at 2nd equivalence point. Given $K_1 = 7.5 \times 10^{-4}$, $K_2 = 10^{-8}$, $K_3 = 10^{-12}$
(A) $\sim 10^{-4}$ (B) $\sim 10^{+4}$ (C) $\sim 10^{-7}$ (D) $\sim 10^{-12}$

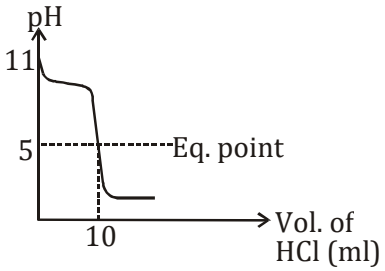
CIE286

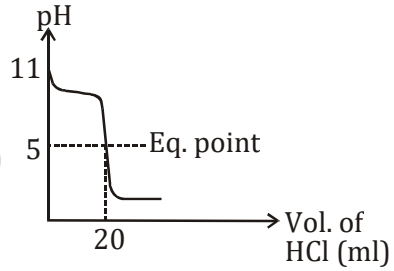
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7. 20 ml of 0.2M RNH_2 ($K_b = 10^{-5}$) is titrated against 0.2 M HCl. Which of the following graph is correct-

(A)


(B)


(C)


(D)


CIE295

8. Best indicator to detect end point in complete titration of 0.2 M K_2CO_3 against 0.1M HCl is –
[For H_2CO_3 : $K_{a_1} = 4 \times 10^{-7}$; $K_{a_2} = 5 \times 10^{-11}$]

(A) Phenolphthalein
(B) Thymol blue ($pK_{In} \simeq 2$)
(C) Methyl orange
(D) Malachite green ($pK_{In} \simeq 1$)

CIE294

9. Select correct statement for 50ml 0.1M H_2A(aq.) solution; $K_{a_1} = 10^{-5}$; $K_{a_2} = 10^{-8}$

(A) $[H^+] = 2[A^{2-}]$
(B) pH of 0.1M H_2A solution is 3
(C) In above H_2A solution when 5 milimoles of NaHA are added then pH increases by 2 units
(D) 50 ml of 0.1M NaOH required to nuetralised completely 50 ml of 0.1M H_2A solution

CIE299

Match the column :

10. On titration of 5 ml, 0.8 M sodium carbonate solution with 0.4N HCl :

Column-I (Volume of HCl added in ml)	**Column-II** (For H_2CO_3 $K_{a_1} = 10^{-7}, K_{a_2} = 10^{-11}$)
(A) 0	(P) Buffer solution of HCO_3^- and CO_3^{2-}
(B) 5	(Q) pH = 7
(C) 10	(R) pH = 9
(D) 15	(S) pH < 7
	(T) $[OH^-] > [H^+]$

CIE289

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	C	B	A	B	C	D	C	BC	(A) → (T), (B) → (P,T), (C) → (R,T), (D) → (Q)

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RACE
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Ionic Equilibrium : RACE-04

Solubility Equilibria of sparingly soluble salts, Solubility and Solubility product constant (K_{sp})

1. A solution of $Pb(NO_3)_2$ is added drop wise to a second solution in which $[Cl^-] = [F^-] = [I^-] = \left[SO_4^{2-}\right] = 0.001$ M. What is the first precipitate that forms ?

(A) $PbCl_2(K_{SP} = 1.5 \times 10^{-5})$ (B) $PbF_2[K_{SP} = 3.7 \times 10^{-8}]$

(C) $PbI_2(K_{sp} = 8.5 \times 10^{-9})$ (D) $PbSO_4(K_{SP} = 1.8 \times 10^{-8})$

CIE265

2. A sparingly soluble salt gets precipitated only when the product of concentration of its ions in the solution (Q_{sp}) becomes greater than its solubility product. If the solubility of $BaSO_4$ in water is 8×10^{-4} mol dm^{-3}. Calculate its solubility in the 0.01 mol dm^{-3} of H_2SO_4 solution

CIE267

3. Calculate the volume of water required to dissolve 0.1 g lead (II) chloride to get a saturated solution. (K_{sp} of $PbCl_2 = 3.2 \times 10^{-8}$, atomic mass of Pb = 207u)

CIE270

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Simultaneous solubility; Solubility of A_2B; AB_2 type of salt and solubility of complex salts

4. What is the pH of a saturated solution at 25°C of milk of magnesia $Mg(OH)_2$ if the K_{SP} of this compound is 4×10^{-12} ?

(A) 3.5 (B) 9.2 (C) 10.3 (D) 10.9

CIE266

5. The solubility of $Mg(OH)_2$ in a buffer of pH = 10 is found to be 0.0232 gm/lit. Molar solubility of $Mg(OH)_2$ in pure water would be (in mol/lit)

(A) 10^{-5} (B) 3.2×10^{-8} (C) 3.2×10^{-5} (D) 10^{-4}

CIE275

6. The salt $Al(OH)_3$ involved in the following two equilibria,

$Al(OH)_3(s) \rightleftharpoons Al^{3+}(aq.) + 3OH^-(aq.)$, K_{sp}

$Al(OH)_3(s) + OH^-(aq) \rightleftharpoons Al(OH)_4^-(aq.)$, K_c

Which of the following relationship is correct at which solubility is minimum?

(A) $\left[OH^-\right] = \left(\frac{K_{sp}}{K_c}\right)^{1/3}$ (B) $\left[OH^-\right] = \left(\frac{K_c}{K_{sp}}\right)^{1/4}$

(C) $\left[OH^-\right] = \sqrt{\left(\frac{K_{sp}}{K_c}\right)^{1/4}}$ (D) $\left[OH^-\right] = \left(\frac{3K_{sp}}{K_c}\right)^{1/4}$

CIE282

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7. A saturated solution of silver benzoate, $C_6H_5CO_2Ag$ has a pH of 8.63. K_a for benzoic acid is 6.5×10^{-5}. Estimate the value of K_{sp} of silver benzoate.

(A) 1.49×10^{-3} (B) 1.399×10^{-2}

(C) 1.287×10^{-2} (D) None of these

CIE283

8. The solubility product of Al $(OH)_3$ is 2.7×10^{-11}. Calculate its solubility in gL^{-1} and also find out pH of this solution. (Atomic mass of Al = 27 u) (at 25°C)

CIE268

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ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	D	(6.4×10^{-5})	0.18	C	D	D	B	$(7.8 \times 10^{-2}, 11 + \log 3)$

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